

Project Title

Full-Stack Deployment & Project Architecture – E-Commerce Product Catalog

Objective

The objective of this project is to design, develop, optimize, and deploy a professional E-Commerce Product Catalog web application using modern frontend web technologies. The project demonstrates modular application architecture, client-side routing, responsive UI design, asset optimization, and live deployment on modern hosting platforms.

Tools Used

- HTML5
- CSS3
- JavaScript
- Visual Studio Code
- Netlify / Vercel Deployment Platform
- Google Chrome Browser

Dataset Description

This project uses static product information including product names, prices, categories, and images to simulate an E-Commerce product catalog system.

Machine Learning Algorithms Used

No machine learning algorithms are used in this project because it focuses on frontend architecture, responsive web design, and deployment concepts.

Key Features

- Modular Frontend Architecture

- Responsive Product Catalog UI
- Client-Side Routing Navigation
- Dynamic Product Cards
- Optimized Responsive Design
- Professional E-Commerce Layout
- Search & Navigation System
- Product Category Sections
- Live Deployment Ready Structure
- Mobile-Friendly Design

Steps Performed

1. Designed the webpage structure using semantic HTML5.
2. Created a professional responsive UI using CSS3.
3. Developed modular product catalog sections.
4. Implemented client-side routing using JavaScript.
5. Added dynamic navigation functionality.
6. Optimized webpage layout and assets for performance.
7. Created responsive product cards and category layouts.
8. Tested responsiveness across devices.
9. Prepared project for deployment on Netlify/Vercel.

Output

- Successfully developed a professional E-Commerce Product Catalog application.
- Implemented modular frontend architecture and routing.
- Added responsive and optimized UI design.
- Created deployment-ready project structure.
- Improved user experience with seamless navigation.

Professional HTML, CSS & JavaScript Code

```
<!DOCTYPE html>  
<html lang="en">
```

```
<head>

<meta charset="UTF-8">

<meta name="viewport"
content="width=device-width, initial-scale=1.0">

<title>E-Commerce Product Catalog</title>

<style>

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:'Segoe UI',sans-serif;
}

body{

    background:#f4f7fb;

    color:#333;
}

/* Header */

header{

    background:#0f172a;

    color:white;

    padding:20px 40px;

    display:flex;

    justify-content:space-between;

    align-items:center;

    flex-wrap:wrap;
```

```
}

header h1{

    color:#38bdf8;
}

nav{

    display:flex;

    gap:20px;
}

nav a{

    color:white;

    text-decoration:none;

    font-weight:bold;

    transition:0.3s;
}

nav a:hover{

    color:#38bdf8;
}

/* Hero Section */

.hero{

    height:350px;

    background:
        linear-gradient(rgba(0,0,0,0.5),
            rgba(0,0,0,0.5)),
        url('https://images.unsplash.com/photo-1523275335684-
37898b6baf30');
}
```

```
    background-size:cover;

    background-position:center;

    display:flex;

    justify-content:center;

    align-items:center;

    color:white;

    text-align:center;
}

.hero h2{

    font-size:50px;
}

/* Product Grid */

.container{

    padding:40px;
}

.products{

    display:grid;

    grid-template-columns:
    repeat(auto-fit,minmax(250px,1fr));

    gap:25px;
}

/* Product Card */

.card{

    background:white;
```

```
border-radius:20px;

overflow:hidden;

box-shadow:0 8px 20px rgba(0,0,0,0.1);

transition:0.3s;
}

.card:hover{

    transform:translateY(-8px);
}

.card img{

    width:100%;

    height:220px;

    object-fit:cover;
}

.card-content{

    padding:20px;
}

.card-content h3{

    margin-bottom:10px;
}

.price{

    color:#2563eb;

    font-size:22px;

    font-weight:bold;
```

```
        margin:10px 0;
    }

    button{

        width:100%;

        padding:12px;

        border:none;

        border-radius:10px;

        background:#2563eb;

        color:white;

        font-size:16px;

        cursor:pointer;

        transition:0.3s;
    }

    button:hover{

        background:#1d4ed8;
    }

    /* Footer */

    footer{

        background:#0f172a;

        color:white;

        text-align:center;

        padding:20px;

        margin-top:40px;
```

```
}

/* Client Side Pages */

.page{

    display:none;
}

.active{

    display:block;
}

/* Responsive */

@media(max-width:768px){

    header{

        flex-direction:column;

        gap:15px;
    }

    .hero h2{

        font-size:35px;
    }

    .container{

        padding:20px;
    }
}

</style>

</head>

<body>
```



```
<header>
```

```
  <h1>ShopSphere</h1>
```

```
  <nav>
```

```
    <a href="#home"
      onclick="showPage('home')">
```

```
      Home
```

```
    </a>
```

```
    <a href="#products"
      onclick="showPage('products')">
```

```
      Products
```

```
    </a>
```

```
    <a href="#about"
      onclick="showPage('about')">
```

```
      About
```

```
    </a>
```

```
    <a href="#contact"
      onclick="showPage('contact')">
```

```
      Contact
```

```
    </a>
```

```
  </nav>
```

```
</header>
```

```
<!-- Home Page -->
```

```
<section id="home"
  class="page active">
```

```
<div class="hero">

    <h2>
        Modern E-Commerce Catalog
    </h2>

</div>

</section>

<!-- Products Page -->

<section id="products"
class="page">

<div class="container">

    <h2 style="margin-bottom:25px;">
        Featured Products
    </h2>

    <div class="products">

        <div class="card">

            <img src=
"https://images.unsplash.com/photo-1542291026-7eec264c27ff">

            <div class="card-content">

                <h3>Running Shoes</h3>

                <p>
                    Premium sports running shoes
                    with modern design.
                </p>

                <div class="price">
                    ₹2999
                </div>

            </div>

        </div>

    </div>

</div>
```

```
<button>
  Add to Cart
</button>
```

```
</div>
```

```
</div>
```

```
<div class="card">
```

```
  <img src=
  "https://images.unsplash.com/photo-1526170375885-4d8ecf77b99f">
```

```
  <div class="card-content">
```

```
    <h3>Smart Camera</h3>
```

```
    <p>
      High-quality DSLR camera
      for photography.
    </p>
```

```
    <div class="price">
      ₹45999
    </div>
```

```
    <button>
      Add to Cart
    </button>
```

```
  </div>
```

```
</div>
```

```
<div class="card">
```

```
  <img src=
  "https://images.unsplash.com/photo-1511707171634-5f897ff02aa9">
```

```
  <div class="card-content">
```

```
    <h3>Smart Phone</h3>
```

```
<p>
    Latest generation smartphone
    with powerful performance.
</p>

<div class="price">
    ₹69999
</div>

<button>
    Add to Cart
</button>

</div>

</div>

</div>

</div>

</section>

<!-- About Page -->

<section id="about"
class="page">

<div class="container">

    <h2>About Us</h2>

    <br>

    <p>

        ShopSphere is a modern E-Commerce
        platform developed using HTML,
        CSS, and JavaScript with modular
        frontend architecture and
        responsive design principles.
```

```
        </p>

</div>

</section>

<!-- Contact Page -->

<section id="contact"
class="page">

<div class="container">

    <h2>Contact Us</h2>

    <br>

    <form>

        <input type="text"
        placeholder="Enter Name"
        style="width:100%;
        padding:12px;
        margin-bottom:15px;">

        <input type="email"
        placeholder="Enter Email"
        style="width:100%;
        padding:12px;
        margin-bottom:15px;">

        <textarea rows="5"
        placeholder="Enter Message"
        style="width:100%;
        padding:12px;"></textarea>

        <br><br>

        <button type="submit">
            Send Message
        </button>
```

```
        </form>

</div>

</section>

<footer>

    <p>
        © 2026 ShopSphere E-Commerce Catalog
    </p>

</footer>

<script>

/* Client Side Routing */

function showPage(pageId){

    const pages =
    document.querySelectorAll(".page");

    pages.forEach(page => {

        page.classList.remove("active");

    });

    document.getElementById(pageId)
    .classList.add("active");

}

</script>

</body>
</html>
```

Conclusion

This project successfully demonstrates full-stack project architecture concepts by developing a professional E-Commerce Product Catalog web application. The application includes modular frontend architecture, responsive design, client-side routing, optimized layouts, and deployment-ready structure suitable for modern hosting platforms.

Expected Outcome

- Fully functional E-Commerce Product Catalog
- Responsive professional UI design
- Client-side seamless navigation
- Optimized and deployment-ready architecture
- Improved frontend project structuring skills
- Live deployment support using Netlify/Vercel

OUTPUT

ShopSphere

HomeProductsAboutContact

Modern E-Commerce Catalog

© 2026 ShopSphere E-Commerce Catalog

ShopSphere

HomeProductsAboutContact

Featured Products



Running Shoes

Premium sports running shoes with modern design.

₹2999

Add to Cart



Smart Camera

High-quality DSLR camera for photography.

₹45999

Add to Cart



Smart Phone

Latest generation smartphone with powerful performance.

₹69999

Add to Cart

© 2026 ShopSphere E-Commerce Catalog

About Us

ShopSphere is a modern E-Commerce platform developed using HTML, CSS, and JavaScript with modular frontend architecture and responsive design principles.

Contact Us

Enter Name

Enter Email

Enter Message

Send Message